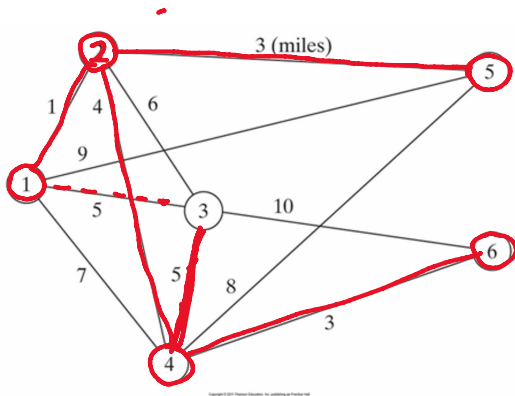




$$C_0 = \emptyset \quad \bar{C}_0 = N = \{1, 2, 3, 4, 5, 6\}$$

$$\checkmark C_1 = \{1\} \quad \bar{C}_1 = \{2, 3, 4, 5, 6\} \quad j^* = 2$$

$(1,2)$, $(1,3)$, $(1,4)$, $(1,5)$, $(1,6) \rightarrow X$.

$$\checkmark C_2 = \{1, 2\} \quad \bar{C}_2 = \{3, 4, 5, 6\} \quad j^* = 5$$

$(1,3)$, $(1,4)$, $(1,5)$, $(1,6)$, $(2,3)$, $(2,4)$, $(2,5)$, $(2,6)$.

$$\checkmark C_3 = \{1, 2, 5\} \quad \bar{C}_2 = \{3, 4, 6\} \quad j^* = 4$$

$$C_4 = \{1, 2, 4, 5\}$$

$$\bar{C}_4 = \{3, 6\} \quad j^* = 6$$

$$C_5 = \{1, 2, 4, 5, 6\}$$

$$\bar{C}_5 = \{3\} \quad j^* = 3$$

$$C_6 = N$$

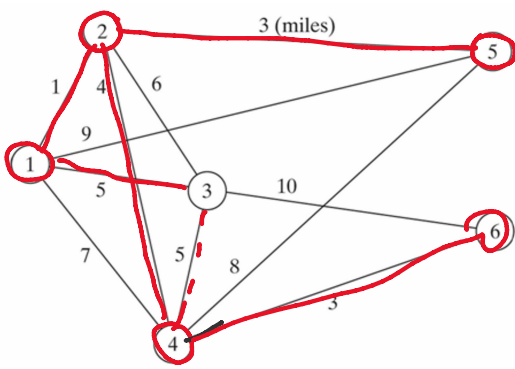
$$\bar{C}_6 = \emptyset$$

$$C_0 = \emptyset \quad \bar{C}_0 = N$$

$$C_1 = \{5\} \quad \bar{C}_1 = \{1, 2, 3, 4, 6\} \quad j^* = 2$$

$$C_2 = \{2, 5\} \quad \bar{C}_2 = \{1, 3, 4, 6\} \quad j^* = 1$$

$$C_3 = \{1, 2, 5\} \quad \bar{C}_3 = \{3, 4, 6\} \quad j^* = 4$$

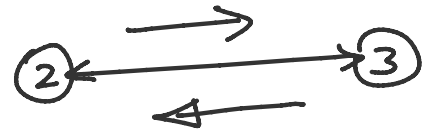
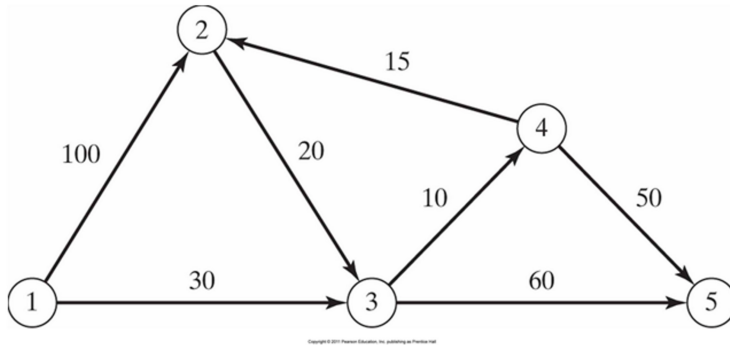
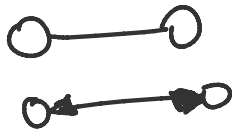


$$C_4 = \{1, 2, 4, 5\} \quad \bar{C}_4 = \{6, 3\} \quad i=6$$

$$C_5 = \{1, 2, 4, 5\} \quad \bar{C}_5 = \{3\}$$

$$C_6 = \emptyset$$

$$\bar{C}_6 = \emptyset$$



	<u>Node</u>	<u>Label</u>	<u>Status</u>
$i=1$	1	$[0, -]$	P
	2	$[0+100, 1]$	T
	3	$[0+30, 1]$	T $\Rightarrow$ P.
$i=3$	1	$[0, -]$	P
	2	$[100, 1]$	T.
	3	$[30, 1]$	P
	4	$[30+10, 3]$	T $\Rightarrow$ P.
	5	$[30+60, 3]$	T.
	1	$[0, -]$	P
	2	$[100, 1]$ or $[40+5, 4]$	T
	3	$[30, 1]$	P
	4	$[40, 3]$	P
	5	alternative routes: $[90, 3]$ or $[40+50, 4]$	T
	2	$[55, 4]$	P.

1		
2	[55, 4]	P.
3	[30, 1] X	P
4	[40, 3]	P
5	[90, 3] or [90, 4] (T) # $\Rightarrow$ P.	

Source 1.

Shortest route between 1-2.

(2) $\rightarrow$ [55, 4] $\rightarrow$ (4) $\rightarrow$ [40, 3] $\rightarrow$ (3) $\rightarrow$ [30, 1] $\downarrow$
~~X~~(1)

$\rightarrow$ 1 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 2 with a shortest length of 55.

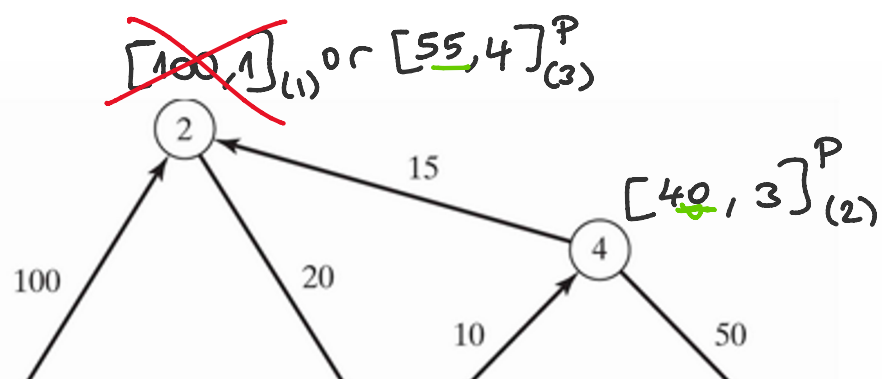
Shortest route between 1-5

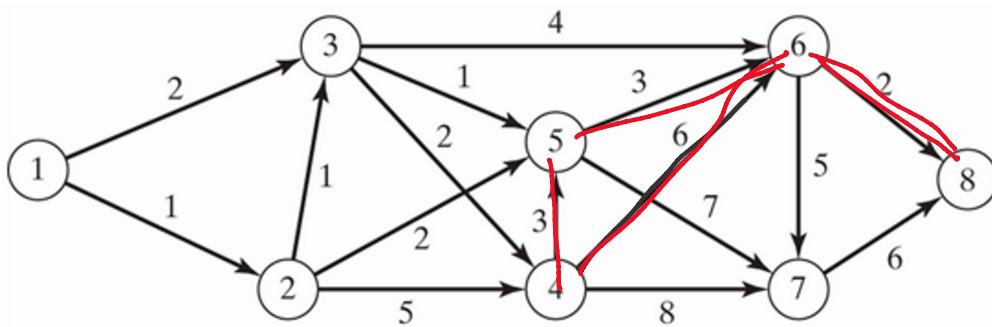
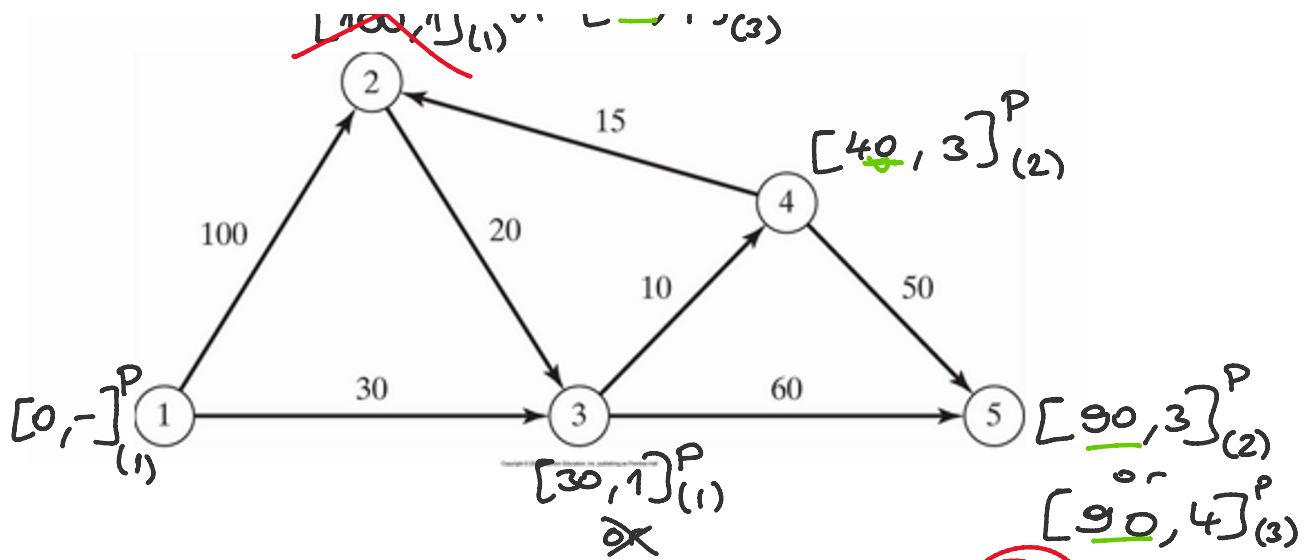
5 $\rightarrow$ [90, 3] $\rightarrow$ 3 $\rightarrow$ [30, 1] $\rightarrow$ 1 stop.

$\rightarrow$ [90, 4] $\rightarrow$ 4 $\rightarrow$ [40, 3] $\rightarrow$ 3 $\rightarrow$ [30, 1] $\rightarrow$ 1 stop.

90: 1 $\rightarrow$ 3 $\rightarrow$ 5
 miles

90: 1 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5





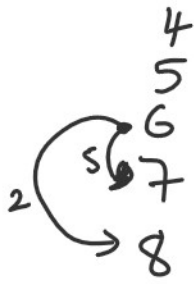
c) $4 \rightarrow 8$.

Source = 4.

Node	Label	Status
4	$[0, -]$	P
5	$[3, 4]$	T *
6	$[6, 4]$	T
7	$[8, 4]$	T

4	$[0, -]$	P
5	$[3, 4]$	P
6	alt. results. $[6, 4]$ or $[6, 5]$	T *
7	$[8, 4]$ or $[10, 5]$	T

4	$[0, -]$	P
5	$[3, 4]$	P



$[0, -]$
 $[3, 4]$
 $[6, 4]$ or $[6, 5]$
 all $[8, 4]$ or ~~$[6+5, 6]$~~
 $[8, 6]$

P
 P
 P
 T ~~W~~
 T



$[0, -]$
 $[3, 4]$
 $(6, 4)$ or $(6, 5)$
 $(8, 4)$ or $(8, 6)$
 $[8, 6]$ or ~~$[8+6, 7]$~~

P
 P
 P
 P
 T $\Rightarrow$ P

$4 \rightarrow 8 : \text{length} = 8$

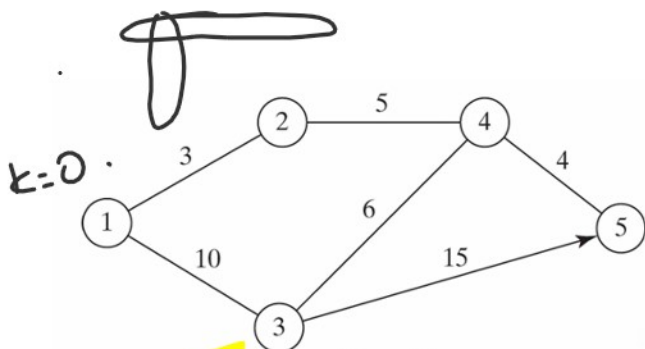
Shortest route:

$8 \rightarrow [11, 6] \rightarrow [6, 4] \rightarrow 4$ stop.

$\rightarrow [6, 5] \rightarrow 5 \rightarrow [3, 4] \rightarrow 4$ stop.

$4 \rightarrow 6 \rightarrow 8$

$4 \rightarrow 5 \rightarrow 6 \rightarrow 8$



$n = 5$

	1	2	3	4	5
1	-				
2		-			
3			-		
4				-	
5					-

Do:

	1	2	3	4	5
1	-	3	10	∞	∞
2	3	-	∞	5	∞
3	10	∞	-	6	15
4	∞	5	6	-	4
5	∞	∞	∞	4	-

$k=1$

$S_0 =$

	1	2	3	4	5
1	-	2	3	4	5
2	1	-	3	4	5
3	1	2	-	4	5
4	1	2	3	-	5
5	1	2	3	4	-

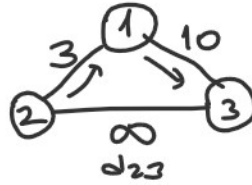
k

3 $\xrightarrow{1}$ 10

$3 + \infty ? < 5$

$$3 + 10 < \infty$$

$$d_{21} + d_{13} < d_{23}$$



$$3 + \infty ? < 5$$

$$D_1 =$$

	1	2	3	4	5
1	-	3	10	∞	∞
2	3	-	13	5	∞
3	10	13	-	6	15
4	∞	5	6	-	4
5	∞	∞	∞	4	-

$k = 2$

$$S_1 =$$

	1	2	3	4	5
1	-	2	3	4	5
2	1	-	1	4	5
3	1	1	-	4	5
4	1	2	3	-	5
5	1	2	3	4	-

$$D_2 =$$

	1	2	3	4	5
1	-	3	10	8	∞
2	3	-	13	5	∞
3	10	13	-	6	15
4	8	5	6	-	4
5	∞	∞	∞	4	-

$k = 3$

$S_2 =$

	1	2	3	4	5
1	-	2	3	2	5
2	1	-	1	4	5
3	1	1	-	4	5
4	2	2	3	-	5
5	1	2	3	4	-

$$D_3 =$$

	1	2	3	4	5
1	-	3	10	8	25
2	3	-	13	5	28
3	10	13	-	6	15
4	8	5	6	-	4
5	∞	∞	∞	4	-

$k = 4$

$S_3 =$

	1	2	3	4	5
1	-	2	3	2	3
2	1	-	1	4	3
3	1	1	-	4	5
4	2	2	3	-	5
5	1	2	3	4	-

$$D_4 =$$

	1	2	3	4	5
1	-	3	10	8	12
2	3	-	11	5	9
3	10	11	-	6	10
4	8	5	6	-	4
5	12	9	10	4	-

$k = 5$

$S_4 =$

	1	2	3	4	5
1	-	2	3	2	4
2	1	-	4	4	4
3	1	4	-	4	4
4	2	2	3	-	5
5	4	4	4	4	-

No change.

$k = 6 = n + 1$ (Stop)